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Fireblocks Yubikey authentication

Updated 8 months ago

Important

If you use or change to YubiKey authentication for the workspace Owner, all users added to the workspace afterward will be required to use YubiKey authentication as well.

Overview

Fireblocks requires a strong method of authentication for transaction signing and authorization. The Fireblocks mobile app uses a device's biometric sensor for authentication by default.

For customers seeking an alternative authentication method, Fireblocks allows customers to use YubiKey.

This article describes the steps to setting up a YubiKey 5 NFC device for authentication on the Fireblocks mobile app. The procedure involves initializing the YubiKey device and sending a list of parameters to Fireblocks to associate the initialized device with a workspace user.

Setting up your YubiKey

1. Download and install the YubiKey personalization tool from the [YubiKey website](#).

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Use the YubiKey Personalization Tool to configure the two slots on your YubiKey on Windows, macOS, and Linux operating systems. The tool works with any YubiKey (except the Security Key). You can also use the tool to check the type and firmware of a YubiKey, or to perform batch programming of a large number of YubiKeys. In addition, you can use the extended settings to specify other features, such as to disable fast triggering, which prevents the accidental triggering of the nano-sized YubiKeys when only slot 1 is configured.

(This tool works on all Yubico devices except the FIDO U2F Security Key, the Security Key by Yubico, and the Security Key NFC by Yubico.)

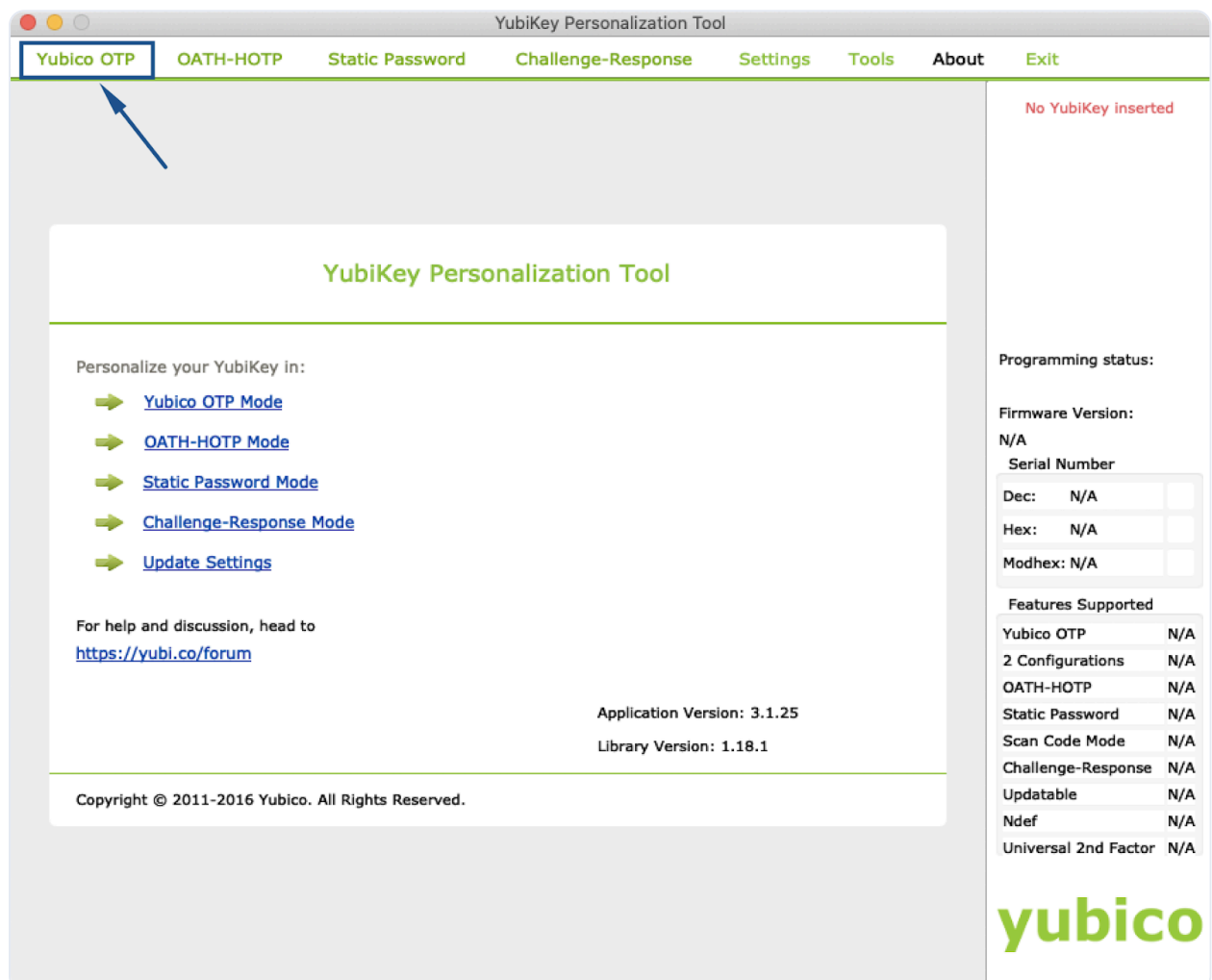
YubiKey Personalization Tool (preferred) v3.1.25

- [Linux Download](#)
- [Mac Download \(.pkg file\)](#)
- [Microsoft Windows Download](#)

YubiKey Personalization Tool (command line interface) v1.19.0

- [Linux Download](#)
- [Mac Download](#)
- [Microsoft Windows 64-Bit Edition Download](#)
- [Microsoft Windows 32-Bit Edition Download](#)

2. Open the YubiKey personalization tool app, then select the **Yubico OTP** tab.



3. Select **Advanced**.

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QUICK

Quickly program a YubiKey for use with Yubico Validation Server

Advanced

Allows you to program one or more YubiKeys with greater control over the configuration values

Programming status:

Firmware Version: N/A

Serial Number

Dec: N/A

Hex: N/A

Modhex: N/A

Features Supported

Yubico OTP	N/A
2 Configurations	N/A
OATH-HOTP	N/A
Static Password	N/A
Scan Code Mode	N/A
Challenge-Response	N/A
Updatable	N/A
Ndef	N/A
Universal 2nd Factor	N/A



4. Select **Configuration Slot 1**.

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5. Select **Generate** next to Public Identity, Private Identity, and Secret Key.

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Select the configuration slot to be programmed

☒ Configuration Slot 1 ☐ Configuration Slot 2

☐ Program Multiple YubiKeys

☐ Automatically program YubiKeys when inserted

Parameter Generation Scheme
Identity from serial; Randomize Secrets

Configuration Protection (6 bytes Hex)
YubiKey(s) unprotected - Keep it that way

Current Access Code
☐ Use Serial Number

New Access Code
☐ Use Serial Number

Yubico OTP Parameters

☒ Public Identity (1-16 bytes Modhex)

Public Identity Length (6 bytes is default length as required by Yubico OTP validation server)

☒ Private Identity (6 bytes Hex)

Secret Key (16 bytes Hex)

Actions
Press Write Configuration button to program your YubiKey's selected configuration slot

Results

#	Public Identity (Modhex)	Status	Timestamp

Programming status:

Firmware Version: N/A

Serial Number

Dec: N/A

Hex: N/A

Modhex: N/A

Features Supported

Yubico OTP	N/A
2 Configurations	N/A
OATH-HOTP	N/A
Static Password	N/A
Scan Code Mode	N/A
Challenge-Response	N/A
Updatable	N/A
Ndef	N/A
Universal 2nd Factor	N/A



6. Verify your YubiKey is inserted into the computer USB slot, then select **Write Configuration**.

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Select the configuration slot to be programmed

☒ Configuration Slot 1 ☐ Configuration Slot 2

☐ Program Multiple YubiKeys

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Parameter Generation Scheme
Identity from serial; Randomize Secrets

Configuration Protection (6 bytes Hex)
YubiKey(s) unprotected - Keep it that way

Current Access Code
☐ Use Serial Number

New Access Code
☐ Use Serial Number

Yubico OTP Parameters

☒ Public Identity (1-16 bytes Modhex) df tu hd rj vg rj

Public Identity Length
6 (6 bytes is default length as required by Yubico OTP validation server)

☒ Private Identity (6 bytes Hex) e3 83 e9 2e 8b ae

Secret Key (16 bytes Hex) 41 54 e6 08 cd df 48 14 6f ba 2e 93 db 3b f7 a0

Actions
Press Write Configuration button to program your YubiKey's selected configuration slot

Results

#	Public Identity (Modhex)	Status	Timestamp
✓ 1	[REDACTED]	YubiKey has been successfully configured	10/03/2020 17:29

Programming status:
Slot 1 configured
Firmware Version:
5.1.2
Serial Number

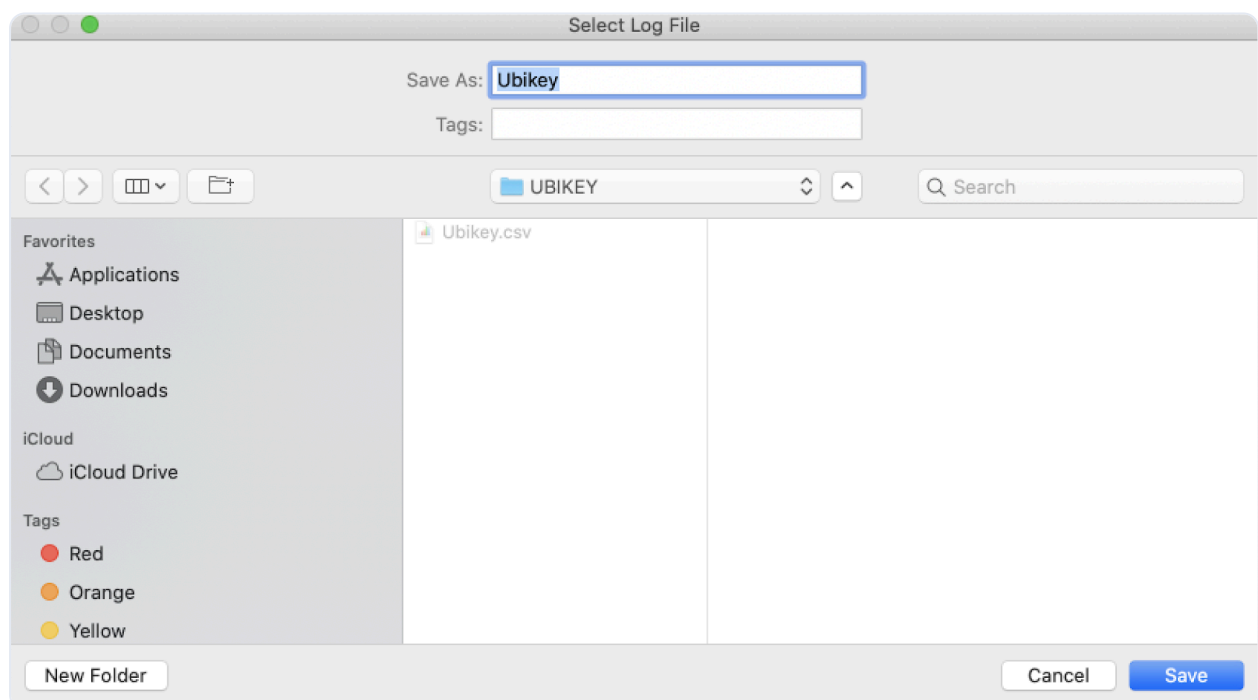
Dec: 9859952
Hex: 967370
Modhex: khieic

Features Supported

Yubico OTP	✓
2 Configurations	✓
OATH-HOTP	✓
Static Password	✓
Scan Code Mode	✓
Challenge-Response	✓
Updatable	✓
Ndef	✓
Universal 2nd Factor	✓

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7. Save as a CSV file.



8. Encrypt the CSV file using [this PGP public key](#) and attach it to your request form.

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warning

It is important that you *DO NOT* send an unencrypted file. If you need assistance with encryption, please let us know or contact your internal IT security team for help.

1. Download gnupg by running one of the following commands on Mac or Linux:

```
brew install gnupg
```

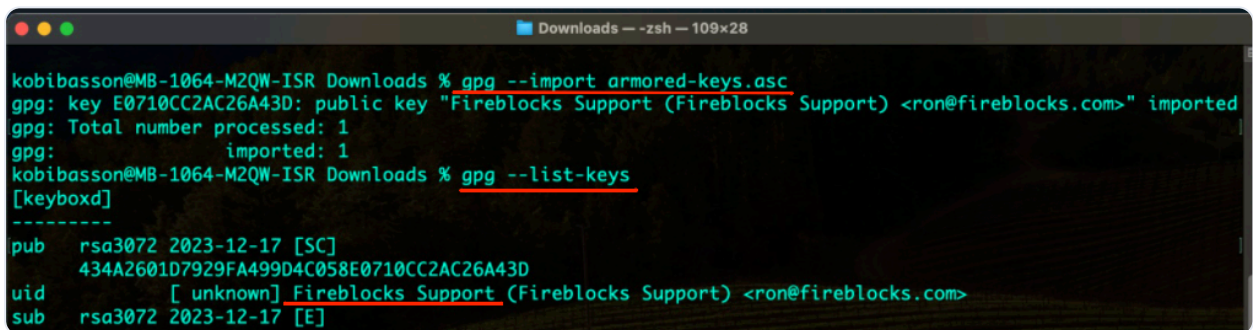
or

```
sudo apt-get install gnupg
```

2. Import the downloaded PGP key using the following commands:

```
gpg --import armored-keys.asc
```

```
gpg --list-keys
```

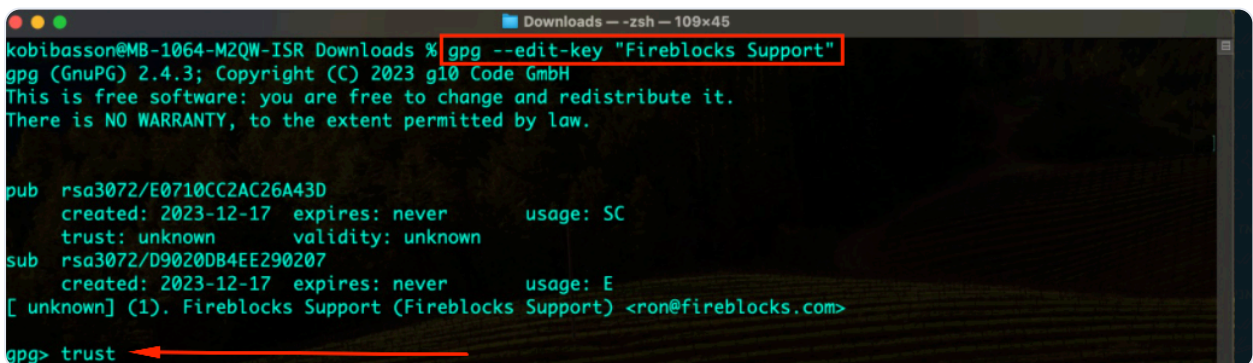


```
kobibasson@MB-1064-M2QW-ISR Downloads % gpg --import armored-keys.asc
gpg: key E0710CC2AC26A43D: public key "Fireblocks Support (Fireblocks Support) <ron@fireblocks.com>" imported
gpg: Total number processed: 1
gpg:      imported: 1
kobibasson@MB-1064-M2QW-ISR Downloads % gpg --list-keys
[keyboard]
-----
pub   rsa3072 2023-12-17 [SC]
      434A2601D7929FA499D4C058E0710CC2AC26A43D
uid   [ unknown] Fireblocks Support (Fireblocks Support) <ron@fireblocks.com>
sub   rsa3072 2023-12-17 [E]
```

3. Edit the trust value to ultimate:

```
gpg --edit-key "Fireblocks Support"
```

```
gpg> trust
```



```
kobibasson@MB-1064-M2QW-ISR Downloads % gpg --edit-key "Fireblocks Support"
gpg (GnuPG) 2.4.3; Copyright (C) 2023 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

pub   rsa3072/E0710CC2AC26A43D
      created: 2023-12-17  expires: never       usage: SC
      trust: unknown      validity: unknown
sub   rsa3072/D9020DB4EE290207
      created: 2023-12-17  expires: never       usage: E
[ unknown] (1). Fireblocks Support (Fireblocks Support) <ron@fireblocks.com>

gpg> trust
```

4. Enter **5** and **y** to confirm.

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```

S = I trust ultimately
m = back to the main menu

Your decision? 5
Do you really want to set this key to ultimate trust? (y/N) y

pub  rsa3072/E0710CC2AC26A43D
    created: 2023-12-17  expires: never       usage: SC
    trust: ultimate      validity: unknown
sub  rsa3072/D9020DB4EE290207
    created: 2023-12-17  expires: never       usage: E
[ unknown] (1). Fireblocks Support (Fireblocks Support) <ron@fireblocks.com>
Please note that the shown key validity is not necessarily correct
unless you restart the program.

gpg> quit
kobibasson@MB-1064-M2QW-ISR Downloads %

```

(Exit the gpg editing window by typing **quit** and selecting **Enter** on your keyboard)

5. You can verify the trust value by running the `gpg --list-keys` command:

```

kobibasson@MB-1064-M2QW-ISR Downloads % gpg --list-keys
[keyboard]
-----
pub  rsa3072 2023-12-17 [SC]
    434A2601D7929FA499D4C058E0710CC2AC26A43D
uid  [ultimate] Fireblocks Support (Fireblocks Support) <ron@fireblocks.com>
sub  rsa3072 2023-12-17 [E]
kobibasson@MB-1064-M2QW-ISR Downloads %

```

6. Run the `gpg -e -r "Fireblocks Support" [filename].csv` command to encrypt the .csv file that you received from the Yubico Personalization Tool.

7. Run the `ls` command to confirm a .gpg file is created.

Supported YubiKeys

We support the following YubiKeys, which you can purchase:

For iOS and Android

- YubiKey 5C NFC
- YubiKey 5 NFC
- YubiKey 5C NFC FIPS
- YubiKey 5 NFC FIPS.

For Android only

- YubiKey 5Ci
- YubiKey 5Ci FIPS

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